

File Handling in Java Demonstration

Title: File Handling – Reading and Writing Files in Java and Finding the Frequency of Patterns

Objective/Aim:

To write the given text into a file named "sample.txt", read the file, and find the frequency of the patterns "pe" and "pi".

System Requirements :

Hardware

- Standard PC/Laptop[cite: 1]
- Intel Core i3/i5 processor or equivalent[cite: 1]
- Minimum 4 GB RAM[cite: 1]
- Minimum 500 MB free storage[cite: 1]
- Keyboard and Monitor[cite: 1]

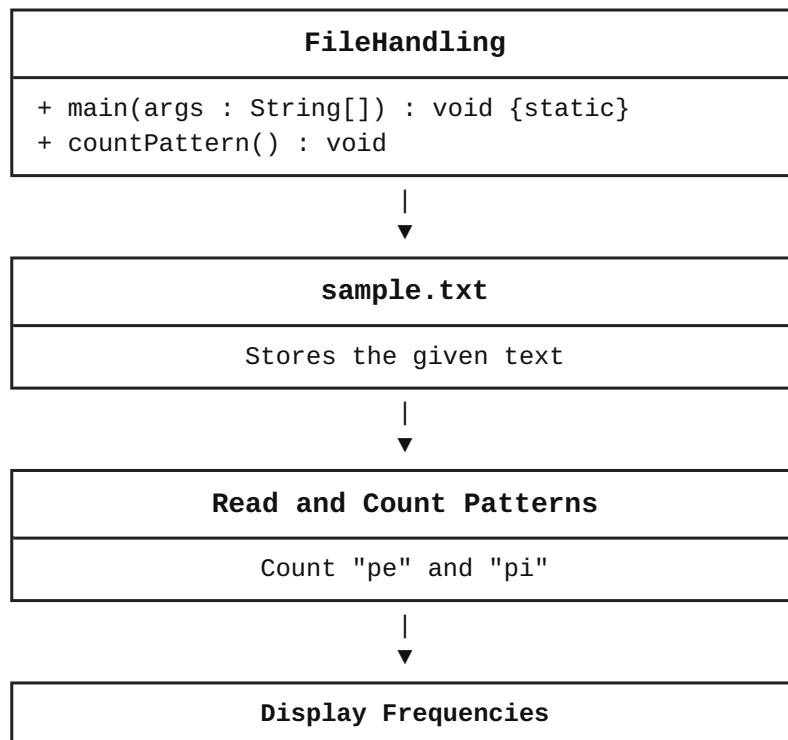
Software

- Operating System: Windows/Linux/macOS[cite: 1]
- Java Development Kit (JDK 17 or JDK 21)[cite: 1]
- Visual Studio Code / Eclipse IDE or any Java IDE[cite: 1]
- Command Prompt/Terminal[cite: 1]

Algorithm / Logic Description:

1. Start the program.
2. Create a file named `sample.txt`.
3. Write the given text into the file using `FileWriter`.
4. Close the file after writing.
5. Read the contents of the file using `FileReader` and `BufferedReader`.
6. Count the occurrences of the patterns "pe" and "pi".
7. Display the frequency of both patterns.
8. Stop the program[cite: 2].

UML class Diagram



Relationship: `FileHandling` interacts with the file system by writing raw text to `sample.txt`, reading it back sequentially via buffered streams, and computing substring frequencies.

Source Code:

```
import java.io.*;

public class FileHandling {
    public static void main(String[] args) {
        String text = "Peter Piper picked a peck of pickled peppers\n"
            + "A peck of pickled peppers Peter Piper picked\n"
            + "If Peter Piper picked a peck of pickled peppers\n"
            + "Where's the peck of pickled peppers Peter Piper picked?";

        try {
            // Write the text into sample.txt
            FileWriter fw = new FileWriter("sample.txt");
            fw.write(text);
            fw.close();

            // Read the contents of sample.txt
            BufferedReader br = new BufferedReader(
                new FileReader("sample.txt"));

            StringBuilder content = new StringBuilder();
            String line;

            while ((line = br.readLine()) != null) {
                content.append(line).append("\n");
            }
            br.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

```

        // Count occurrences of pe and pi
        String data = content.toString().toLowerCase();
        int peCount = 0;
        int piCount = 0;

        for (int i = 0; i < data.length() - 1; i++) {
            String pattern = data.substring(i, i + 2);

            if (pattern.equals("pe")) {
                peCount++;
            }

            if (pattern.equals("pi")) {
                piCount++;
            }
        }

        System.out.println("'pe' - no of occurrences - "
            + peCount);
        System.out.println("'pi' - no of occurrences - "
            + piCount);

    } catch (IOException e) {
        System.out.println("An error occurred: "
            + e.getMessage());
    }
}
}

```

Compilation & Execution Commands:

Save file as: FileHandling.java

Compilation: javac FileHandling.java

Execution: java FileHandling

Output Screenshot:

```

PS C:\Users\mmeen\RAVI> javac FileHandling.java
PS C:\Users\mmeen\RAVI> java FileHandling
'pe' - no of occurrences - 20
'pi' - no of occurrences - 12
PS C:\Users\mmeen\RAVI>

```

Result:

Thus, the given text was successfully written into "sample.txt", read using Java file handling operations, and the frequencies of the patterns "pe" and "pi" were counted and displayed.